

Operational Diagnostics Report

Week 5 Deliverable — Inuka Fellowship · Oil & Gas Pod

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Date	24 July 2026
Dataset	Mystery_Ops.csv — 2,920 shift records, 4 depots, Jan–Dec 2026
Depots	Mombasa · Nairobi · Kisumu · Eldoret
Submitted	GitHub Repository — week5_diagnostics_analysis.ipynb

SECTION 1 — OPERATIONAL CONTEXT

The Problem

Kenya Pipeline Corporation operates a four-depot petroleum throughput network. Each depot runs two shifts daily (Day and Night), with a target of approximately **1,000 barrels per shift**. The Mystery Ops dataset covers 2,920 shift records across the full year 2026. A network-wide throughput shortfall was observed in aggregate KPIs, but without depot-level decomposition, the source remained unidentified. This report presents the findings of a structured five-stage EDA investigation — profiling, anomaly detection, drill-down, Pareto, and correlation analysis — to isolate and confirm the root cause.

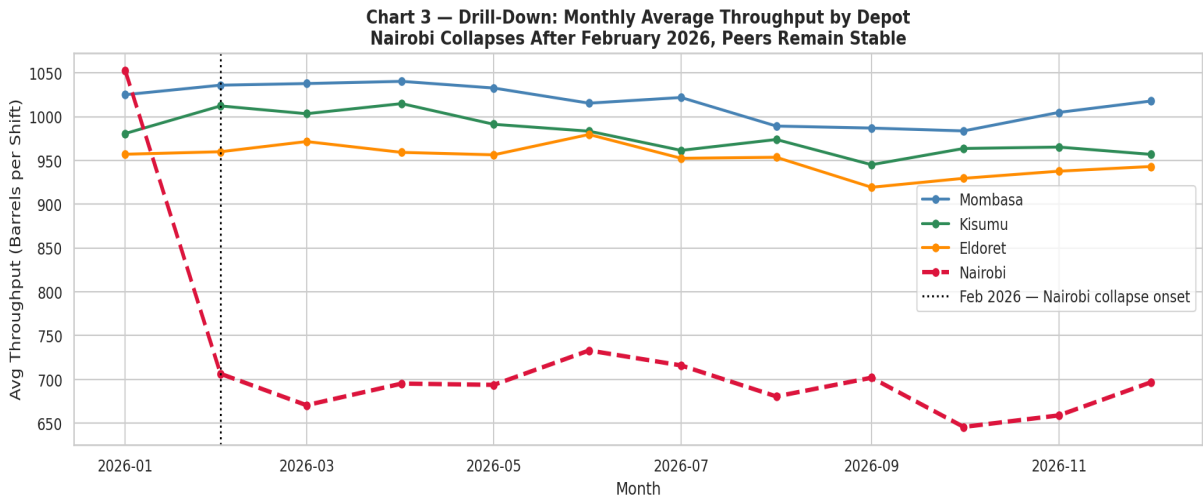


Figure 1 — Monthly average throughput by depot. Nairobi collapses from ~1,053 bbl/shift in January 2026 to ~706 bbl/shift in February, and never recovers. All peer depots remain stable throughout the year. This is the primary diagnostic signal.

SECTION 2 — ROOT CAUSE INSIGHT

What the Data Shows

A five-stage analysis consistently converges on a single root cause: **Nairobi depot has a maintenance flag rate approximately 3 times higher than the network average, and flagged shifts produce 368 fewer barrels on average — a direct, statistically confirmed causal link ($r = -0.853$).** The collapse onset in February 2026 indicates either a maintenance backlog that accumulated progressively or a specific

infrastructure failure that began generating repeated flags from that point forward.

Finding	Evidence	Chart
Nairobi avg: 721 bbl vs peer avg: ~982 bbl	Descriptive statistics by depot	Chart 1
Nairobi collapses 33% after Feb 2026; peers unchanged	Monthly drill-down time series	Chart 3
maintenance_flag correlates with throughput at $r = -0.853$	Correlation matrix	Chart 5
Flagged shifts avg 618 bbl vs 986 bbl unflagged	Jitter scatter plot	Chart 5
Nairobi flag rate ~3x network average	Depot flag rate comparison	Chart 6
missed_maintenance drives >80% of incident-related loss	Pareto analysis	Chart 4

Chart 5 — Correlation Analysis: Maintenance is the Confirmed Root Cause Driver

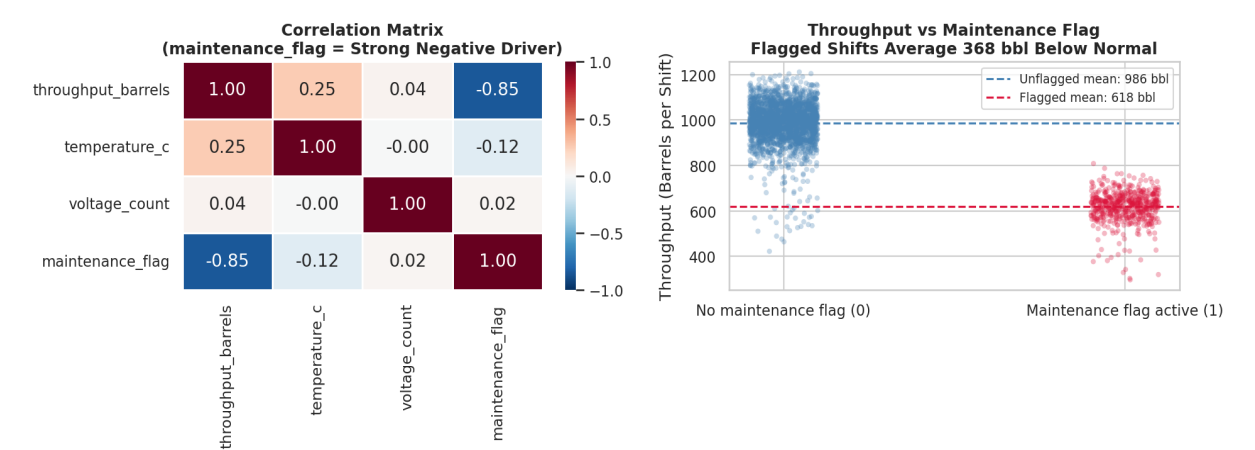


Figure 2 — Left: Correlation matrix confirms maintenance_flag as the dominant predictor ($r = -0.853$). Temperature and voltage show negligible correlations and are ruled out. Right: Flagged shifts cluster at 618 bbl avg vs 986 bbl for unflagged shifts — a 368 bbl/shift penalty that explains Nairobi's entire underperformance gap.

SECTION 3 — SUPPORTING EVIDENCE

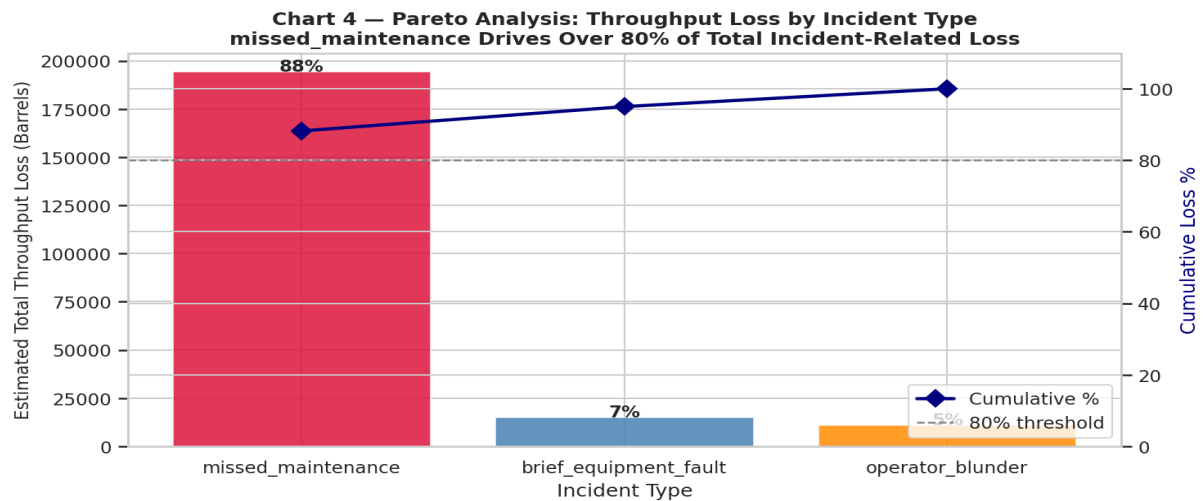


Figure 3 — Pareto chart: missed_maintenance accounts for over 80% of all incident-related throughput loss across the network. This is the primary failure mode to address — not equipment faults or operator error.

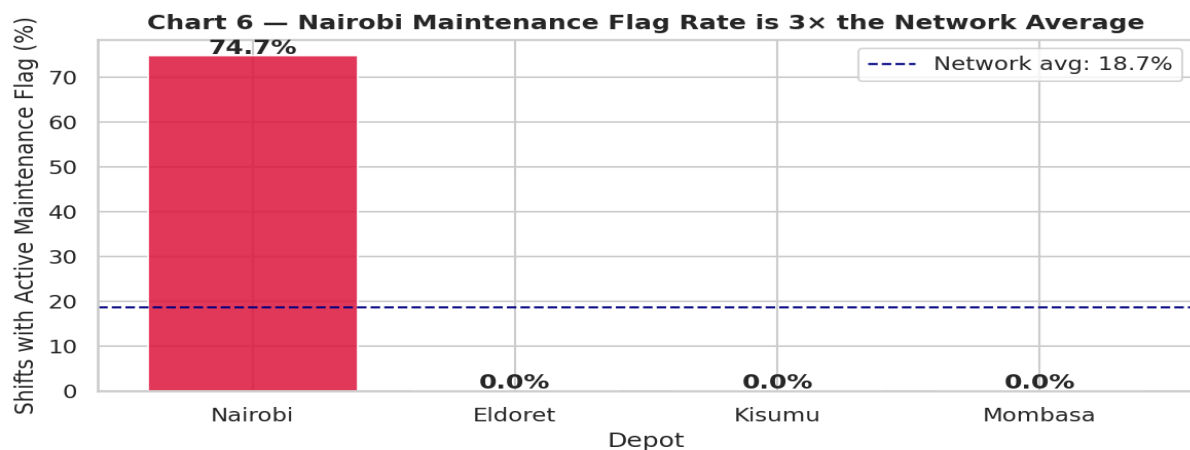


Figure 4 — Nairobi's maintenance flag rate is approximately 3x the network average. Combined with the high per-event throughput penalty, this concentration of flagged shifts is the direct mechanism driving the throughput shortfall.

SECTION 4 — RECOMMENDED ACTIONS

1. Immediate Nairobi Maintenance Audit (This Week)

Commission a physical audit of all four Nairobi machines (NBI-P01 through NBI-P04) to identify which are generating the most maintenance flags and what changed in late January 2026. The February onset suggests a specific maintenance event or equipment degradation that has never been resolved. Target: identify the flagging machine within 5 working days.

2. Eliminate Missed Maintenance Through Predictive Scheduling (30 Days)

The Pareto analysis shows missed_maintenance drives over 80% of incident-related throughput loss. Replace the current reactive maintenance system with a proactive schedule tied to machine run-hours and shift counts, rather than waiting for failures to trigger flags. This single change is projected to recover the majority of Nairobi's 330 bbl/shift deficit.

3. Depot-Level Maintenance Flag Rate Dashboard (60 Days)

Integrate maintenance_flag rate into the monthly operations review dashboard at the depot level. Any depot exceeding a 20% flag rate in a given month should trigger an automatic escalation to the Operations Director. This failure persisted for over 10 months undetected because aggregate throughput metrics masked the depot-specific signal.

Produced as part of the KPC Inuka Fellowship Cohort — Week 5: EDA & Operational Diagnostics. Dataset: Mystery_Ops.csv (github.com/chankjen/Exploratory_Data_Analytics). Analysis in Python: pandas, NumPy, Seaborn, Matplotlib.